Paper 9 — Which ReLU Saddles Can We Escape?

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Loss Landscape of Shallow ReLU-like Neural Networks: Stationary Points, Saddle Escape, and Network Embedding

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The paper in one minute

Question: How do we tell local minima from escapable saddles when ReLU losses have kinks and the ordinary gradient may not exist?
Model/Method: One-hidden-layer ReLU-like networks, squared error, and one-sided directional derivatives. Define *escape neurons* through first-order directional conditions.
Finding: No escape neuron implies a local minimum. For a scalar output, an escape neuron is also sufficient to rule out a local minimum. Small living neurons account for saddle escape in the analyzed vanishing-initialization setting.
Why we care: Geometry at nonsmooth stationary points helps explain plateaus, saddle transitions, and what wider networks preserve.

1. First understand the network and the ReLU kink

The authors study a **one-hidden-layer** network. Each hidden neuron takes an inner product with the input, applies a ReLU-like activation, and passes its contribution to the output:

$$\textbf{Model: } \hat{y}(x) = H\rho(Wx), \quad \rho(z) = \alpha^{+}z \ (z \geq 0),\ \alpha^{-}z \ (z < 0).$$

The loss is the empirical squared prediction error. Unlike a smooth function, ReLU changes slope abruptly when a hidden weight becomes orthogonal to a training input: $\mathbf{w_i} \cdot \mathbf{x_k} = \mathbf{0}$. At such a kink, there need not be one well-defined gradient.

2. A tiny example: directional derivatives at a kink

For **f(z)=|z|**, there is no two-sided derivative at 0, but moving a small amount in either direction increases the value. At **g(z)=−|z|**, moving either way decreases it. The missing derivative alone does not tell us whether the point is safe or escapable.

$$D_d f(0) = \lim_{t \to 0+} [\ f(td) - f(0)\ ] / t = |d| \geq 0.$$

Directional stationarity (Definition 3.5)
A point P is **directionally stationary** when every one-sided directional derivative is nonnegative. It cannot be improved to *first order* in any direction; it may still have a second-order decreasing direction. This is the notion used throughout the paper, rather than Clarke stationarity.

3. The big insight: an escape neuron identifies a hidden descent path

The authors split changes in a hidden input weight into a **radial** part (change its length) and a **tangential** part (change its direction). At a directionally stationary point, ordinary output/radial derivatives vanish, but tangential one-sided derivatives need only be nonnegative.

Definition 4.1, in words. A hidden neuron is an *escape neuron* if some tangential direction has zero current loss derivative, yet a relevant underlying coefficient is nonzero. This means that jointly changing the hidden direction and its output weight can unlock descent that is invisible to a simple first-order test.

4. Worked example: a stationary saddle with a hidden escape route

A pedagogical one-neuron ReLU example, compatible with the paper's shallow/scalar-output setting (not a numerical experiment from the authors):

$$x = (1,0), \quad y = 1, \quad \hat{y} = a \cdot \mathrm{ReLU}(w \cdot x), \quad L(a,w) = (1/2)(\hat{y} - 1)^2.$$

Step 1. Start at a=0, w=(0,0). The prediction is zero and L=1/2. A perturbation of size t to either the output weight alone or the input weight alone cannot change the prediction.

Step 2. Any joint perturbation of all parameters proportional to t changes the output by order t^2, so every first-order directional derivative at this point is zero. It is directionally stationary.

Step 3. Perturb both weights together: a=t and w=(t,0), with t>0. The neuron turns on, predicts t^2, and:

$$L(t,(t,0)) = (1/2)(1 - t^2)^2 = 1/2 - t^2 + (1/2)t^4 < 1/2.$$

So the loss decreases at **second order**, even though it cannot decrease to first order. This is the escape mechanism: **the two weights must move together**. For this one-neuron example, the zero output weight and the nonzero directional coefficient meet Definition 4.1.

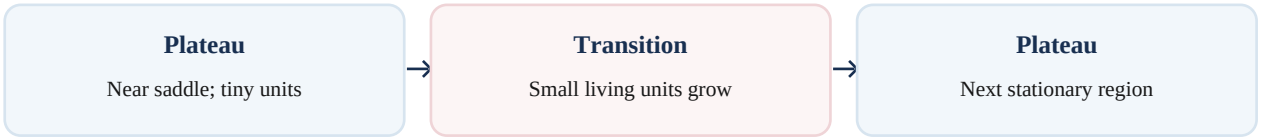
> **Why the math matters**
> The paper's Corollary 4.4 formalizes this pattern for any **non-minimum stationary point with one output**: some direction reduces loss by $C\delta^2$ at leading order, with C<0. This is a loss-decreasing direction, not a universal claim that the standard Hessian always exists.

5. Theorem 4.2 — the exact logical statement

Setting	What is actually proved
Any output dimension	No escape neurons implies local minimum.
Scalar output only	Local minimum if and only if no escape neurons. Equivalently: escape neuron implies not a local minimum.
Several outputs	An escape neuron does **not** by itself prove the point is not a local minimum.

6. Why does gradient descent pause, then suddenly improve?

In the very-small-initialization regime, gradient flow may approach a sequence of non-minimum stationary points. While the network lingers nearby, the loss barely decreases; when small, still-active neurons grow, the loss falls more rapidly.



Fact 4.5: in the scalar-output ReLU gradient-flow model, each neuron conserves the difference between squared input-weight norm and squared output weight. This is why tiny, balanced initialization keeps their amplitudes tied together:

$$\frac{d}{dt}\left(\|w_i\|^2 - a_i^2\right) = 0; \quad \text{at vanishing balanced initialization, } \|w_i\| \approx |a_i|.$$

Corollary 4.6: for the specified scalar-output gradient-flow trajectory with initialization scaled toward zero, any reachable non-minimum stationary point must contain at least one hidden neuron whose **input and output parameters are both zero**.

Fact 4.7: a strictly loss-decreasing path from a stationary point must change escape-neuron parameters. Consequently, the analyzed saddle escapes use **amplitude growth of small living neurons**, not only rearrangement of previously large units.

Dead neuron vs small living neuron
A **dead** ReLU neuron is inactive on all training inputs and does not create an escape path under the local perturbations considered. A **small living** neuron still responds to data; its input/output weights can amplify together and drive the loss away from a nearby saddle.

7. What Figure 3 actually shows

The authors train a 50-hidden-neuron ReLU network on **five synthetic samples** from tiny Gaussian initialization. The run has three plateaus; two end around **42,000** and **152,000** epochs, while the final plateau is identified as a local minimum. Curves for neuron directions, input-weight norms, and output weights show aligned groups whose amplitudes grow during escapes.

Do not overinterpret: this experiment illustrates the theorem and proposed training narrative. It is *not* a general convergence-rate bound or evidence that every real neural network follows this sequence.

8. Scope of the dynamical conclusion

The strongest saddle-to-saddle statements here are for **one-hidden-layer, scalar-output ReLU networks under gradient flow and vanishing initialization**. The general stationary-point classification covers ReLU-like networks and allows multiple outputs, but the equivalent local-minimum criterion does not.

9. Network embedding: making a wider net represent the same function

A narrow model can be represented inside a wider one. Because ReLU is positively homogeneous, one neuron can be copied into two while preserving the predictions.

$$a \cdot \mathrm{ReLU}(w \cdot x) = (a/2) \cdot \mathrm{ReLU}(w \cdot x) + (a/2) \cdot \mathrm{ReLU}(w \cdot x).$$

More generally, split one neuron into copies with input weights $\beta_l w$ and output weights $\gamma_l a$, choosing $\beta_l > 0$ and $\sum_l \beta_l \gamma_l = 1$. The input-output function is unchanged, **but the parameter-space geometry can change**.

> **Proposition 4.8: preservation conditions**
> For unit replication of a stationary neuron, stationarity is preserved if and only if **all its tangential derivatives vanish** or **every γ_l is nonnegative**. A type-1 local minimum is preserved if and only if all those tangential derivatives vanish or **every γ_l is strictly positive**. Merely preserving network predictions is not enough.

10. Main contributions, assumptions and limitations

Aspect	What to remember
Main contributions	Directional stationarity handles ReLU kinks; escape-neuron criterion; second-order escape path; embedding preservation conditions.
Core assumptions	One hidden layer; ReLU-like piecewise-linear activation; empirical squared loss. Dynamic corollary: scalar output, gradient flow, vanishing initialization.
Critical limitations	No general deep-network theorem; no universal finite-time saddle-escape rate; converse of Theorem 4.2 not proved for multiple outputs; no guarantee of global minima.

11. Cross-paper connections for our survey

Lee (2016): smooth strict-saddle avoidance; **Jin (2017):** finite-time perturbed-GD stationarity; **Zhang (2026):** how saddles organize progressive learning in several architectures; **Wu (2025):** how to recognize escapable stationary points despite ReLU nonsmoothness, and why neuron growth is required in its particular dynamical regime.

12. Quick reference

Concept	Short meaning
Directional stationary	One-sided loss derivative is nonnegative in every direction.
Escape neuron	Hidden unit satisfying Definition 4.1's first-order tangential criterion.
Theorem 4.2	No escape neuron implies local min; converse for scalar output.
Corollary 4.4	Scalar-output non-minimum stationary point has an $O(\delta^2)$ descent path.
Proposition 4.8	Unit replication preserves stationarity/minimality only under specific conditions.

Papers Master — brief note

Wu et al. (2025) characterize directional stationary points of shallow ReLU-like squared-loss networks and define escape neurons. No escape neurons ensures a local minimum (iff for scalar output). Under vanishing-initialization scalar-output gradient flow, saddle escape requires growth of small living neurons; unit replication need not preserve stationarity.

Reading priority: paper pp. 3–10, especially Definitions 3.5 and 4.1, Theorem 4.2, Corollary 4.6, Figure 3, and Proposition 4.8. The examples above are pedagogical adaptations. Original: ICLR 2025, Wu–Simsek–Ged.